

Geography Ch-2 (Advanced) — Geological Structure of India

Advanced / India-applied | GS-I & GS-III | Source: D.R. Khullar (India: A Comprehensive Geography)
Complements Must-Do Ch-2 (Rocks) with India's rock systems & mineral link
Extra CA: National Critical Mineral Mission / GSI Deccan exploration

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

ADVANCED — India's Rock Systems & Mineral Geology (D.R. Khullar)

■ NEWS TRIGGER

The National Critical Mineral Mission (NCMM, launched 2025) tasks GSI with 1,200 exploration projects (2024–2031); in 2025-26 GSI mapped >22,000 sq km of the Deccan Traps and used magnetotellurics + drone magnetic surveys to target rare earths, lithium and cobalt.

■ INTRO & ORIGIN

India's rocks fall into four great systems by age — **Archaean (Dharwar)**, **Purana (Cuddapah–Vindhyan)**, **Dravidian (Gondwana/Palaeozoic–Mesozoic)** and **Aryan (Deccan Trap & Tertiary)** — each tied to specific minerals and landforms.

The **Peninsular block** is built of ancient Archaean crystalline rocks (granite, gneiss, schist); the **extra-Peninsula (Himalaya)** is young, folded sedimentary rock — the framework behind India's mineral distribution.

■ KEY DATA

Rock system	Age	Economic / geographic note
Archaean / Dharwar	Oldest (crystalline, metamorphosed)	Metallic minerals: iron, manganese, gold (Kolar)
Cuddapah–Vindhyan (Purana)	Proterozoic	Building stone, limestone, diamonds (Panna)
Gondwana	Upper Palaeozoic–Mesozoic	~98% of India's COAL; iron ore
Deccan Trap	Cretaceous–Eocene basalt	Black cotton (regur) soil; basalt building stone
Tertiary / Siwalik	Cenozoic	Petroleum; Himalayan structure

■ STATIC THEORY

- The **Archaean/Dharwar** crystallines form the Peninsular shield and host most metallic minerals (iron, manganese, gold).
- **Gondwana rocks** hold nearly all of India's coal reserves — a direct geology→energy link.
- The **Deccan Trap** is one of the world's largest flood-basalt provinces (~500,000 sq km); its weathering yields fertile black cotton (regur) soil.
- Weathering of ancient crystalline (granite/gneiss/schist) rocks rich in ferromagnesian minerals produces the red soils of the Peninsula.
- The Himalaya is built of young Tertiary sedimentary strata — structurally distinct from the rigid Peninsular block.

■ MUST-KNOW FACTS

1. Four systems: Archaean (Dharwar), Purana (Cuddapah–Vindhyan), Dravidian (Gondwana), Aryan (Deccan Trap–Tertiary).
2. Gondwana rocks = ~98% of India's coal.

■ UPSC TRAPS

- ✗ The Deccan Trap is a sedimentary formation.
- ✓ The Deccan Trap is IGNEOUS (flood basalt / extrusive volcanic rock).

3. Deccan Trap basalt → black cotton (regur) soil. 4. Dharwar crystallines → iron, manganese, gold (Kolar). 5. Panna (Vindhyan) → diamonds; Peninsular shield = oldest, most stable. 6. NCMM 2025 → GSI targeting rare earths / lithium in Deccan & Western India.	✗ Himalayan rocks are ancient crystallines. ✓ Himalayan rocks are young TERTIARY sedimentary strata; the Peninsula holds the ancient crystallines. ✗ Most Indian coal is in the Deccan Trap. ✓ Almost all Indian coal is in GONDWANA rocks, not the Deccan Trap.
■ MAINS ANGLE <i>Link India's geological structure to its mineral wealth and the National Critical Mineral Mission — why Deccan/Peninsular geology is central to India's clean-energy resource security.</i>	■ STUDY LINK Geography → India → Geological Structure & Rock Systems (Khullar Ch-2)

End of Register Notes — UPSC Agent / Copilot CLI